# Kabadiwala Connect — Project Record

Smart India Hackathon 2026 · Problem statement SIH26229 · Ministry of Mines (Clean & Green Technology)

This document records everything the team built, step by step, in simple language: what the app does, how it works inside, the tools we used, the problems we hit and how we solved them. It is updated after every build phase.

Last updated: 24 October 2026, 11:30 pm. **Final hackathon build (version 1.0.0)**, all phases done, tested on a real phone.

## At a glance: final status

| Area | Status |
|---|---|
| Core demo flow: photo → category → weight → spoken price → recyclers → QR handover → recycler confirms → ledger → sync | ✓ Done, works offline, tested on a real phone |
| All plan phases 0–13 | ✓ Done (Phase 12b admin price editor replaced by Firestore console + scripts) |
| Languages | ✓ 12 (Hindi, Garhwali, Marathi, English, Bengali, Bhojpuri, Punjabi, Urdu, Gujarati, Tamil, Telugu, Kannada) |
| On-phone AI scrap recogniser | ✓ 80% on held-out photos, suggests only when 70%+ sure |
| Real data | ✓ 8 authorised recyclers (CPCB list), 7 of 8 prices from published rates, Uttarakhand facts |
| Automatic tests | ✓ 136 passing, `flutter analyze` clean |
| Release | ✓ Signed with the ByteStorms key, uploaded to Firebase App Distribution (1.0.0) |
| App size | ! 34.6 MB for most phones (target 30 MB); 28.7 MB for older phones |
| Still to do by the team | Field study (check prices with local dealers), native-speaker check of translations, more training photos for batteries and motors |

## 1. The problem in one page

India produces a huge amount of electronic waste (old phones, TVs, wires, batteries, circuit boards). Most of it is collected by informal scrap collectors called **kabadiwalas**. They usually:

- sell to the nearest local dealer at a low price, because they don't know the fair rate;
- have no record of what they sold, to whom, or how much they were paid;
- sometimes break or burn the waste (for example burning cables for copper), which is dangerous for their health and the environment;
- often cannot read well, use cheap Android phones, and have weak or no internet.

Meanwhile, **authorised recyclers** (registered with the government to recycle e-waste safely) want more material but can't reach these collectors.

## 2. Our solution

**Kabadiwala Connect** is an Android app that works **without internet** and **without needing to read**. A collector can:

1. Take a photo of the scrap.
2. Pick what it is by tapping a picture (circuit board, copper wire, battery...).
3. Enter the weight with big + and − buttons.
4. **Hear** a fair price spoken aloud in their own language.

5 See nearby **authorised** recyclers, ranked by best rate, distance and pickup.

6 Hand over the scrap and get a **QR code** — a tamper-proof digital receipt.

7 The recycler scans the QR, checks it, and records the payment (cash or UPI).

8 The collector's **earnings ledger** updates. Everything syncs when internet comes back.

### Who uses it

| User | What they do in the app |
|---|---|
| Collector (kabadiwala) | Sells scrap, hears prices, finds recyclers, keeps an earnings record |
| Recycler | Scans the collector's QR, confirms weight and payment, sets their rates |
| Admin (our team) | Updates the price list and recycler list |

### Design rules we follow (for low-literacy users)

| Rule | Why |
|---|---|
| No screen needs reading: every button = big icon + 1–2 words + sound | Many users can't read comfortably |
| At most 4 main choices per screen; buttons at least 56 dp tall | Easy to tap, not confusing |
| Prices are always spoken, never only shown | Numbers on screen are not enough |
| Colour is never the only signal: green + tick = done, amber + clock = waiting, red + warning = danger | Works for colour-blind users too |
| Works offline after first login | Internet is patchy where collectors work |
| Cash is as important as UPI | Most collectors are paid in cash |
| All text comes from language files, never typed into the code | So we can add languages easily |

## 3. Technology we chose (and why)

| Part | Tool | Why we chose it (simple reason) |
|---|---|---|
| App | **Flutter** (Dart language) | One codebase, fast on cheap Android phones |
| State management | **Riverpod** | Keeps screens updated automatically when data changes |
| Database | **Cloud Firestore** (Firebase) | Works offline: saves on the phone first, uploads later |
| Login | **Firebase Phone Auth** | Collectors log in with their phone number + OTP, no passwords |
| Voice | **Android Text-to-Speech** + recorded clips | Speaks prices; recorded clips for languages Android can't speak |
| Tamper-proof receipt | **SHA-256 hash** + **QR code** | Any change to the record changes the code, so cheating is visible |
| Scripts | **Python** (pandas, firebase-admin) | Turns the team's spreadsheet data into app data |

| Part | Tool | Why we chose it (simple reason) |
|---|---|---|
| Cost | Firebase free "Spark" plan | No money needed for the prototype |

**Minimum phone:** Android 7 (released 2016) or newer. The app works on all newer versions up to Android 16.

## 4. How we set up the computer (Mac)

Before writing any code, we installed the tools. Here is what each one is for:

| Tool | What it does |
|---|---|
| Homebrew | An "app store" for developer tools on Mac |
| Java 17 | Needed by Android's build system |
| Flutter 3.47 | The framework the app is written in (includes Dart) |
| Android SDK 36 + build tools | Turns our code into an Android app (APK) |
| Android NDK | Compiles small native parts used by some plugins |
| Gradle 9.3 | Android's build tool; downloads libraries and builds the APK |
| Firebase CLI | Lets us log in to Firebase and upload security rules from the terminal |
| FlutterFire CLI | Connects the Flutter app to our Firebase project |
| Python virtual environment | A private box for our Python scripts' libraries |
| Xcode Command Line Tools | Apple's compiler; `flutter test` needs it |

The full step-by-step guide for teammates is in `docs/SETUP.md`.

### Problems we hit during setup, and how we fixed them

| Problem | Cause | Fix |
|---|---|---|
| Downloads were very slow (about 200 KB/s) | Slow internet connection | Ran downloads in the background and resumed them; switched to smaller downloads where possible |
| First Android build failed: "Operation timed out" while downloading Gradle | Our internet provider could not reach **one** of GitHub's 4 download servers (185.199.109.133). Java kept picking that one. | Downloaded Gradle with `curl` (which picked a working server), checked its SHA-256 checksum, and placed it where Gradle expects it |
| `flutter test` failed: "CarbonCore/Folders.h file not found" | The Mac's Command Line Tools installation was damaged (a file was missing) | Deleted and reinstalled the Command Line Tools |
| Android NDK download (900 MB) kept stalling | Slow internet; Android's downloader deletes the partial file if stopped | Downloaded it with `curl` in resume mode, verified the official SHA-1 checksum, then installed it manually |
| Wanted minimum Android 6 (API 23) | Current Flutter no longer supports Android 6 | Set the minimum to Android 7 (API 24); almost no phones still run Android 6 |

| Problem | Cause | Fix |
|---|---|---|
| Wikimedia stopped our photo downloads ("too many requests") | We downloaded too fast | Used Wikimedia's standard image size and waited between downloads |

### Firebase setup (done in the browser by the team)

1. Created the Firebase project `kabadiwala-connect` (free Spark plan, no Google Analytics).
2. Created the Firestore database `(default)`, **Standard edition**, location **Mumbai (asia-south1)** — closest to our users, so faster.
3. Turned on **Phone sign-in** and added test phone numbers (fake numbers with a fixed OTP, so we can test without real SMS).
4. Logged in with `firebase login`.
5. Downloaded a **service-account key** (a secret file that lets our Python script write data). It is stored as `scripts/serviceAccount.json` and is **never** uploaded to git.
6. Ran `flutterfire configure`, which registered the Android app with Firebase and generated the connection settings (`firebase_options.dart`, `google-services.json`).

## 5. How the project is organised

| Folder | What's inside |
|---|---|
| `app/` | The Flutter Android app |
| `app/lib/core/` | Shared building blocks: languages, audio, theme, hashing, location, sync |
| `app/lib/data/` | Data models, the code that reads/writes the database, offline seed data |
| `app/lib/features/` | One folder per screen group: onboarding, home, selling, prices, recyclers, handover, ledger, safety, recycler mode |
| `app/assets/` | Language files, sounds, icons, offline seed data |
| `app/test/` | Automatic tests that check the logic is correct |
| `data/` | Spreadsheets (CSV) the team fills from the field study: materials, prices, recyclers, areas |
| `scripts/` | Python scripts that load the spreadsheets into the app and the database |
| `docs/` | Plan, setup guide, this record |
| `firestore.rules` | Security rules: who can read and write what in the database |

## 6. What we built, phase by phase

The plan has 14 phases (0–13). Each phase ends with automatic tests, a code check (`flutter analyze`) and a saved version in git.

### Phase 0 — Setup and scaffolding ✓

**Goal:** an empty app that runs, connects to Firebase, and has the right look.

What we made:

- Created the Flutter app (Android only, package `in.kabadiwalaconnect`).
- Added all the libraries the plan needs (camera, location, voice, QR, database...).
- Created the folder layout above.
- **Big-button theme:** every button at least 56 dp tall, large bold text, high-contrast colours, and three status colours (green, amber, red) that are always paired with an icon.
- A placeholder home screen that shows a **green tick** if Firebase connected or an **amber clock** if not. The app never crashes or blocks if Firebase is missing — it just keeps working.

## Phase 1 — Data layer and seed data ✓

**Goal:** the app knows about materials, prices and recyclers, even with no internet on the very first run.

**The data (spreadsheets in `data/`):**

| File | What it holds |
|---|---|
| `materials.csv` | 8 scrap types: circuit board, copper wire, phone battery, vehicle battery, old TV, motor, mixed plastic, old phone. Each has an icon and a hazard level (none / low / high) |
| `prices.csv` | Price range per kg (low–high), last week's price, and the local dealer's price (for comparing earnings later) |
| `recyclers.csv` | Recycler name, phone, location, whether authorised, registration number, pickup or not, and their rate per material |
| `areas.csv` | A few areas with map coordinates, used if a collector doesn't allow location access |

All rows are clearly marked **placeholder** for now. We will replace them with real data from our field study. We never invent real business names or phone numbers.

**How the data gets into the app:**

- `scripts/csv_to_assets.py` turns the spreadsheets into small files that are **built into the app** (`app/assets/seed/`). This is the offline backup.
- `scripts/seed_firestore.py` uploads the same data to the Firestore database.

**Money and weight are stored as whole numbers:** money in **paise** (Rs 1 = 100 paise) and weight in **grams**. Computers make small rounding errors with decimal numbers (for example 0.1 + 0.2 = 0.30000000000000004), which is unacceptable for money. Whole numbers never have this problem.

**Offline-first reading:** when a screen needs the price list, the app shows the built-in copy **immediately**, then switches to the live database copy as soon as it's available. So the app is never empty and never waits for internet.

**Offline-first writing:** when the collector saves something (a lot, a handover), Firestore saves it **on the phone instantly** and uploads it when internet returns. Every record gets a unique ID made on the phone (a UUID), so even if the same record is sent twice, it never becomes two records.

**Data models** (the "shape" of each record): Material, Price, Recycler, Area, User, Lot (one batch of scrap), Handover, Confirmation, Ledger entry.

**One change from the original plan:** the plan had the recycler update the collector's handover record. But if **both** phones are offline, the recycler can't update a record that hasn't reached the server yet. So

the recycler writes a separate **Confirmation** record with the same ID. The collector's app joins them. This works even when both phones are offline.

**Security rules** (`firestore.rules`, uploaded to Firebase):

| Data | Who can read | Who can write |
|---|---|---|
| Materials, prices, areas | Anyone logged in | Admin only |
| Recyclers | Anyone logged in | Admin; an invited recycler can link their login and edit only their own rates |
| A user's profile | Only that user | Only that user (cannot make themselves admin) |
| Lots, ledger | Only the collector who owns them | Only that collector |
| Handovers | The collector and the receiving recycler | Collector creates it; after that it can **never be changed** |
| Confirmations | The collector and the recycler | Only the recycler who received the scrap |

## Phase 2 — Languages and voice ✓

**Goal:** the app speaks and shows text in the user's language, and switching language is instant.

**Languages (12):**

| Code | Language | How it's spoken | Why it's included |
|---|---|---|---|
| hi | Hindi (default) | Android text-to-speech | Most collectors understand it |
| gbm | Garhwali | Recorded clips (Android can't speak it); Hindi voice until recorded | Local language of Uttarakhand |
| mr | Marathi | Android text-to-speech | Required by the problem statement |
| en | English | Android text-to-speech | For judges and recyclers |
| bn | Bengali | Android text-to-speech | Many collectors in North Indian cities come from West Bengal |
| bho | Bhojpuri | Hindi voice until clips are recorded | Many collectors come from Bihar and East UP |
| pa | Punjabi | Android text-to-speech | Spoken around Roorkee / West UP / Punjab |
| ur | Urdu | Android text-to-speech; the whole screen flips right-to-left | Widely spoken in West UP |
| gu | Gujarati | Android text-to-speech | Large e-waste hubs in Gujarat |
| ta | Tamil | Android text-to-speech | Large e-waste hubs in Tamil Nadu |
| te | Telugu | Android text-to-speech | Large e-waste hubs in Telangana / Andhra |
| kn | Kannada | Android text-to-speech | Bengaluru is a major e-waste city |

**Honesty note:** the developers wrote the translations for every language except English. Each file is marked draft — check with a native speaker, and that check must happen before field use. Garhwali is still a copy of the Hindi text until the team translates it.

**Choosing a language without reading:** the first screen shows only **4 big choices** (Hindi, Garhwali, Marathi + "More languages"), keeping to the 4-choice rule. "More" shows all 12. Each tile says its language's name out loud on the first tap and chooses it on the second. The language can be changed any time with the language (translate) button at the top of the home screen.

Each language has one file, for example `assets/i18n/hi.json`. Text can have slots, e.g. "You will get about {amount} rupees" — the app fills in the number.

**How the app decides how to speak (3 steps):**

1. If a **recorded clip** exists for this sentence in this language, play it. Numbers are built from recorded number clips.
2. Otherwise, if the phone's **text-to-speech** knows the language, speak it.
3. Otherwise, speak it in **Hindi**.

**Speaking numbers from recorded clips (NumberSpeaker):** Hindi-family numbers are irregular (e.g. 21 = "ikkees", 45 = "paintalees"), so we record every number from 0 to 99 plus "hundred", "thousand", "lakh", "rupees", "about". The app breaks any amount into these pieces using the Indian system:

| Amount | Clips played |
|---|---|
| Rs 340 | about · 3 · hundred · 40 · rupees |
| Rs 1,340 | about · 1 · thousand · 3 · hundred · 40 · rupees |
| Rs 1,50,000 | about · 1 · lakh · 50 · thousand · rupees |

**Hidden audio-check screen:** long-press the app title to open a list of every sentence with a play button, so the team can check each language.

`scripts/check_i18n.py`**:** checks that every language file has every sentence, and lists which recorded clips are still missing.

## Phase 3 — Onboarding and home ✓

**Goal:** a first-time user can pick a language, log in, set a PIN and reach the home screen; after that the app opens **offline with just the PIN**.

| Screen | What happens |
|---|---|
| S1 Language | Big buttons with each language's name in its own script. **Tap once to hear it, tap again to choose it** — so nobody has to read |
| S2 Login | Phone number on a big number pad (+91 is fixed), then the 6-digit SMS code. For testing we use Firebase test numbers (no real SMS) |
| S3 PIN | Make a 4-digit PIN (typed twice). Every time the app starts, it asks for the PIN — this works **without internet** |
| S4 Role | Only shown if this phone number was invited by a recycler. Everyone else goes straight in as a collector |
| S5 Home | Four big picture tiles: Sell scrap, Today's rates, My earnings, Safety |

How it works inside:

- **The app remembers the login.** Firebase keeps the signed-in user on the phone, so after the first online login the app never needs internet to open.

- **The PIN is never stored.** Only a SHA-256 fingerprint of it (mixed with the user ID) is saved. A 4-digit PIN is a convenience lock for the phone, not strong security, and we say so.
- **Every screen talks.** A shared screen frame speaks the screen's instruction when it opens, and has a speaker button to repeat it, plus a small **sync icon**: green cloud = everything sent, amber cloud = waiting to send, grey cloud = no internet (the app still works). Tapping the sync icon says what it means.
- **Every big tile** speaks its name when tapped. Long-press only speaks (for "what is this?").
- **Recycler invitation:** a recycler listing in the database has an `ownerPhone`. When that phone number logs in, it may choose recycler mode, and its login is linked to the listing. The security rules check the phone number on the server, so nobody can pretend to be a recycler.
- To make phone login work on Android, we registered the app's signing fingerprints (SHA-1 and SHA-256) with Firebase from the terminal.

## Phase 4 — Selling scrap (photo → category → weight → spoken price) ✓

**Goal:** the heart of the demo. In airplane mode, a collector can photograph scrap, pick what it is, set the weight and **hear a fair price**.

| Screen | What the collector does |
|---|---|
| S6 Photo | Big round camera button, up to 3 photos; a red X on a photo removes it to retake |
| S7 Category | Picture tiles in 2 columns. Dangerous items (batteries, old TVs) have a **red border and warning icon** and say a warning out loud |
| S8 Weight | A huge number with −1, −½, +½, +1, +5 kg buttons. Every change is spoken ("2.5 kilo") |
| S9 Price | Says **"You will get about Rs 300"** and shows the range (Rs 270–Rs 330). The lot is saved immediately, even offline |

**The price formula** (from the plan, all in whole paise):

- low = weight (g) × low rate (paise/kg) ÷ 1000
- high = weight (g) × high rate (paise/kg) ÷ 1000
- spoken price = middle of low and high, rounded to the nearest Rs 10

Example: 1.7 kg circuit board at Rs 180–Rs 220/kg → Rs 306–Rs 374 → **"about Rs 340"**.

**Photos — a decision we had to make:** the plan said to check whether Firebase's photo storage is free. We checked Firebase's official documentation: since **3 February 2026**, free-plan projects cannot use Cloud Storage at all. So we used the plan's backup method:

- the full photo (1280 px, compressed) stays **on the phone**;
- its **SHA-256 fingerprint** is taken the moment it is captured (this goes into the tamper-proof receipt);
- a small thumbnail (480 px, under 200 KB) is saved in Firestore, which **queues it offline** and uploads it when internet returns — no extra upload code needed.

## Phase 5 — Today's rates board ✓

Shows every material with its Rs /kg range and a **trend arrow**: green ↑ (price went up more than 5% since last week), red ↓ (went down more than 5%), grey → (about the same). Tapping a tile says, for

example, "Circuit board: 180 to 220 rupees per kilo. Rate went up." The arrow, colour and spoken word always go together, so colour is never the only signal. Works offline from the saved price list.

## Phase 6 — Finding the best recycler ✓

**Ranking formula** (PLAN.md §6.3), for one material:

| Part | Weight | How it's measured |
|---|---|---|
| Rate | 45% | This recycler's rate ÷ the best rate on the list |
| Distance | 35% | 1 − (distance in km, capped at 25) ÷ 25 |
| Pickup | 20% | 1 if they collect from the collector, else 0 |

- **Unauthorised recyclers are hidden.** The whole point is to move scrap into the formal, safe route.
- Distance is the straight-line distance (haversine formula) — no maps SDK, so it works offline.
- Ties are broken by higher rate, then nearer, then name, so the list never jumps around.
- Each card shows: rank, **authorised** badge, distance, rate per kg, **how much you'll get for this lot**, a big **Call** button and a **Choose** button. Tapping the card reads it all out.

**Location:** the app first explains out loud why it needs location. The collector can tap **My location** (GPS works without internet) or simply **pick their area** from a list if they don't want to share location.

## Phase 7 — Tamper-proof handover with QR, and recycler mode ✓

**Collector side (S11 Receipt):** when the collector chooses a recycler, the app creates a **handover record** (lot, weight, agreed price, time, GPS, photo fingerprints) and seals it with a **SHA-256 hash**:

`v1 | id | collector | recycler | material | weight | price | time | lat | lng | photos-hash` → SHA-256 → 64-character fingerprint

The screen shows this as a **QR code** plus a short reference like **A3F9-1C2B**, which is also spoken one character at a time so it can be read over the phone. A status card shows **amber clock = waiting** until the recycler confirms, then **green tick = confirmed**, with the amount and how it was paid.

**Recycler side (recycler mode):**

| Screen | What the recycler does |
|---|---|
| R1 Home | Three tiles: Scan QR, My rates, Recent deals |
| R2 Scan + verify | Scans the collector's QR. The app **re-computes the hash from the QR's data**: green "Record is genuine", or red "Warning! The record was changed". Then the recycler enters the weight they measured and taps **Cash**, **UPI** or **Later** (money due) |
| R3 My rates | Changes their Rs /kg per material with −10/−1/+1/+10 buttons |

**Why it can't be cheated:** if anyone changes even one digit (for example the weight from 2 kg to 9 kg) the fingerprint no longer matches and the recycler's phone shows a red warning. This needs **no internet**, because the QR carries the whole record.

**Automatic warning flags (rule-based):**

| Flag | When |
|---|---|
| Record was changed | The hash doesn't match |

| Flag | When |
|---|---|
| Weight mismatch | Measured weight differs from the declared weight by more than 10% |
| Price far from market | Agreed price is below 0.7 × the board's low price or above 1.3 × its high price |

**Design change from the plan:** the recycler writes a separate **Confirmation** record instead of editing the collector's handover. This works even when **both** phones are offline, and the collector's original record stays frozen (the security rules forbid changing it).

## Phase 8 — Earnings ledger ✓

**Goal:** the collector always knows what they earned and what they are still owed.

- Three big totals at the top: **Got today**, **Due today**, **This week**; plus **Total still due** shown separately in amber.
- The speaker reads: "Today you got 640 rupees, 200 rupees still due. This week 1,100 rupees."
- Every confirmed handover automatically becomes a ledger entry (with an ID fixed to that handover, so it can never be counted twice).
- **"Sold elsewhere"** button: pick what was sold, type the rupees on the big keypad, tap **Cash** or **UPI**. This keeps the record honest when scrap goes to a local dealer.
- "Today" and "this week" use the phone's **local time** (weeks start on Monday), while everything is stored in UTC.

## Phase 12a — AI that recognises the scrap from the photo ✓

**What it does:** after the collector takes a photo, a small AI model **on the phone** (no internet needed) guesses what the scrap is. If it is at least **70% sure**, that picture tile moves to the top with a green border and a badge like **"Suggested 92%"**, and the app says "Looks like Mixed plastic, 92 percent sure. Tap it if it is right." If it is less sure, it suggests nothing. The collector always makes the final choice.

**How we trained it (on the team's Mac):**

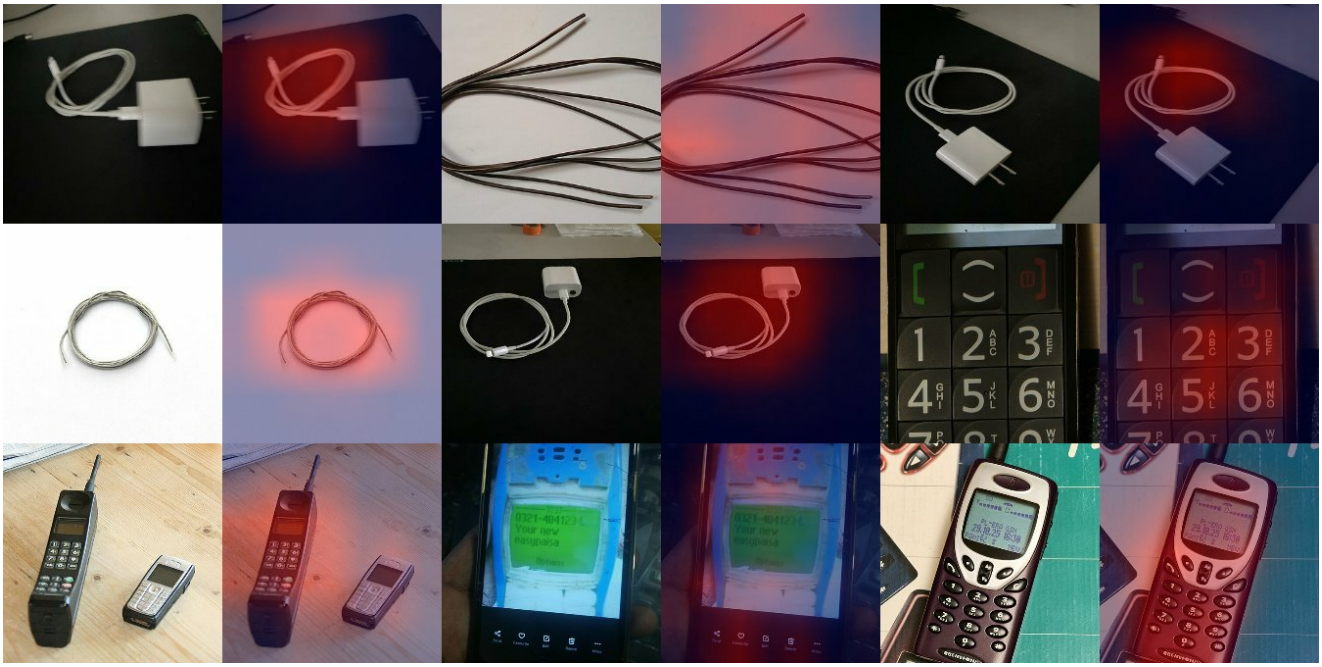
1. **Photos:** 744 photos in 9 folders: the 8 scrap types plus **OTHER** ("not scrap": hands, tables, bags). About 600 are free photos from **Wikimedia Commons** (every author and licence recorded in `data/photos/CREDITS.csv`) and the rest were taken by our team. **We looked at every downloaded photo** and removed about 300 unsuitable ones (people, charts, roofs, power lines).
2. **Model:** MobileNetV2, a small image model made for phones, already trained on millions of general photos. We only taught it our 9 categories ("transfer learning").
3. **Close-up practice:** during training it also sees random **zoomed-in crops**, so it can recognise a phone from just its **camera, ports or buttons**, or a cable from just its end.
4. **Honest test:** 20% of the photos were **held back and never used for training**. The final phone file was tested on them.
5. **Size:** the finished model is **2.7 MB** (int8 TensorFlow Lite).

**Result: 80% correct on 137 photos it had never seen.**

| Category | Accuracy | Category | Accuracy |
|---|---|---|---|
| Phone battery (Li-ion) | 100% | Circuit board | 75% |
| OTHER (not scrap) | 100% | CRT TV | 73% |
| Copper wire / cable | 86% | Vehicle battery (lead-acid) | 57% |
| Mixed plastic | 81% | Motor | 50% |
| Phone | 79% | | |

Weak spots, honestly: lead-acid batteries (only 37 photos) get confused with lithium batteries, and motors (42 photos) are hard. More team photos will fix this. Several plastic photos show the same bottle, so the plastic score may be higher than real-world use.

**What the AI looks at ("heat maps"):** to prove it isn't guessing from the background, we coloured the area of each photo that made the decision. Red = what the AI looked at. It focuses on the **cable coil, the phone keypad and screen**, not the table.



Heat maps: left = photo, right = red where the AI looked. Cables (top), phones (bottom).

## Real data instead of made-up data

**Real authorised recyclers.** We replaced the fake recyclers with the **8 authorised e-waste recyclers of Uttarakhand** from the official **CPCB list (as on 08-06-2023)**, for example M/s Attero Recycling (Bhagwanpur, Roorkee), M/s Root Recycling (Padli Gurjar, Roorkee), M/s Starto Metal Recycle Plant (Roorkee) and M/s Resource E-Waste Solution (Bhadrabad, Haridwar). Names, addresses and registration details are exactly as in the government list. The list gives no phone numbers or rates, so **we did not make any up**: the app shows "Rate not known" and the call button stays off until a verified number is added. Map positions are approximate (village or town centre, from OpenStreetMap).

Because the real recyclers publish no rates, we changed the ranking rule: **authorised recyclers with unknown rates are now shown too**, ranked by distance and pickup only. Unauthorised ones are still always hidden.

**Real scrap prices.** 7 of 8 prices now come from published scrap-rate websites (2025–2026). Each row in `data/prices.csv` records where the number came from:

| Material | Rs per kg | Source (simplified) |
|---|---|---|
| Circuit board | 80 – 200 | recyclinginindia.com, May 2026 |
| Copper wire / cable | 100 – 450 | reuze.in (wiring) and recyclinginindia.com (copper with insulation) |
| Phone battery | 80 – 130 | reuze.in (battery) |
| Vehicle battery | 60 – 120 | todaypricerates.com (lead-acid) |
| CRT TV | 3 – 10 | Rs 50–200 per TV ÷ about 20 kg |
| Mixed plastic | 8 – 14 | recyclinginindia.com; a Roorkee dealer lists Rs 13 |
| Old phone | 125 – 310 | Rs 20–50 per dead phone ÷ about 0.16 kg |
| Motor | 40 – 55 | **Placeholder**: no published rate found |

**Real facts for the presentation** (sources in `data/regional_facts.md`):

| Fact | Figure |
|---|---|
| E-waste generated in India, 2024–25 | **13.97 lakh tonnes** (CPCB, Parliament reply) |
| Uttarakhand e-waste collected/processed, 2024–25 | **1,13,562 tonnes** (Parliament reply) |
| Authorised recyclers in Uttarakhand | **8**, total capacity **1,53,068 tonnes/year** (CPCB list) |
| Share of recycling done by the informal sector | **about 90–95%** (ICRIER study; formal share is rising) |

## Features added after testing on a real phone

We installed the app on a **Motorola Edge 50 Fusion (Android 16)** and tested it live. Based on what we saw, we added:

| Feature | What it does |
|---|---|
| **Logout** | A button on the home screen. It asks "Do you want to log out?" with big Yes / No buttons (spoken). If some data has not been sent yet, it refuses and says "turn on internet first", because logging out then would lose that data. |
| **Back stops the voice** | Pressing Back (button or swipe) stops any speech immediately; the app then waits quietly for the next tap. |
| **Voice on / off switch** | A switch on the home screen, remembered on the phone. When off, the app does not talk by itself, but the speaker button on each screen still reads that screen out when tapped, so a person who can't read can always get help. |
| **Type the weight in kg or grams** | Besides the +/− buttons, a keypad button lets the collector type "350 grams" or "12 kilo". Below 1 kg the app shows and speaks grams. |
| **Live price while setting weight** | The fair price ("~Rs 300") is shown under the weight and on the Next button, and changes as the weight changes. |

**Problems found on the real phone, and fixes:**

| Problem | Fix |
|---|---|
| Login failed: "SMS unable to be sent until this region enabled" | Firebase blocks SMS to every country by default. We allowed **India** in Authentication → Settings → SMS region policy. |

| Problem | Fix |
|---|---|
| Screen titles were white on white (invisible) | The title style had no colour set; we gave it the theme's text colour. |
| The Android build failed after adding the AI library | The library mixed Java 11 and Kotlin 17 settings; we told the build tool to accept that. |
| A row on the weight screen didn't fit a small 360-pixel-wide screen | Caught by an automatic test on a phone-sized screen; we simplified the row. |

## Who can be a recycler: apply and verify

At first the app only offered recycler mode to phone numbers we had linked by hand. After testing, we changed it so that **everyone is asked** after the PIN: **Kabadiwala** or **Recycler**.

| Who taps "Recycler" | What happens |
|---|---|
| A number already linked to an authorised recycler | Recycler mode opens |
| Anyone else | A **"Join as a recycler"** form: pick your business from the **official CPCB list**, or type the name and **registration number**, then **Send application** |
| After sending | "Your application is being checked" , with the option to use the app as a kabadiwala meanwhile |
| After the team approves | "Approved! Open recycler mode" ✓ |

**Why not let anyone be a recycler?** A recycler confirms payments (which become the kabadiwala's earnings record), sets rates shown to everyone, and carries an "Authorised" badge. A cheat could fake all of that. So the **server's security rules** let an applicant save only a pending application, and only for the phone number Firebase verified at login. Only the team can approve, using `scripts/approve_recycler.py`.

We tested this end to end on the real phone with a clearly labelled **fake test listing** ("TEST Recycler (fake - for testing only)", TEST-CPCB-0001): +91 99999 00003 applied, the team approved it, and the phone opened recycler mode.

## Loading screen and app icon

After the PIN, a 2-second **branded loading screen** shows the **logo** (green recycling arrows with a gold Rs coin, "recycle e-waste, get a fair price"), the app name, a tagline ("हर कबाड़ का सही दाम", Fair price for every scrap) and the credit **"Developed by ByteStorms"**, Smart India Hackathon 2026 · SIH26229. A tap skips it. The same logo is the **phone's app icon**, in classic and adaptive versions for Android 8+.



The app icon: classic round (left) and how Android 8+ shows it as a circle or rounded square.

## Phase 9 — Safety guide ✓

The **Safety** tile shows 5 big picture cards with a red "NO" sign or a green tick. Tapping a card reads it aloud:

| Card | Message (simplified) |
|---|---|
| Never burn cables | The smoke is poisonous; strip wires with a tool |
| Don't break old TVs | The picture tube contains lead and can explode; give the whole TV to the recycler |
| Batteries away from fire and water | Swollen or broken batteries can catch fire; don't press or puncture them |
| Wear gloves | Handle circuit boards and broken items with gloves; wash hands after |
| Never use acid | Acid burns and poisons soil and water; sell boards to a recycler |

The **first time** a collector picks a dangerous item (batteries, old TVs), its safety card opens and plays **before** the weight screen. A test checks that every high-hazard material has a card with text in all 12 languages.

## Phase 10 — Sync you can see ✓

The cloud icon on every screen now shows **how many records are waiting** to be sent (a small amber number), and turns green when everything has reached the server. Logout refuses while anything is unsent, so nothing is lost. `docs/offline-test.md` is a printed checklist for testing on a real cheap phone: airplane mode at each step, the app killed mid-sale, the phone restarted with unsent records, two phones both offline, and logout safety.

## Phase 11 — Polish ✓

- **Huge fonts:** many older users set the phone's font to the maximum. We added automatic tests that open every main screen at **2× text size** on a small 360-pixel phone, in Hindi and Tamil. They found 3 screens whose tiles overflowed; we made tile icons and labels **shrink to fit**.
- **Smaller app:** removed an unused 2.5 MB AI graphics library and moved debugging information out of the app.
- **Top bar:** the home screens show the small logo instead of a long title, so the five buttons fit.

## Phase 13 — Demo mode and demo script ✓

- **Demo tools** (long-press the logo on the home screen): show the safety cards again, set the location to Roorkee if indoor GPS is slow, and open the audio check. They never delete records.
- `docs/demo-script.md`: a timed **3-minute demo** with two phones, offline the whole time, switching languages, AI recognition, spoken price, real recyclers, tamper-proof QR, and finally sync. It includes a "what to do if something goes wrong" table.

## Release: a real app, no cables

- **Signing:** release builds are signed with the team's own key ("CN=ByteStorms"). The key and its password are kept **outside git**; the team must keep a private backup, because Android only accepts updates signed with the same key. Its fingerprints are registered with Firebase so phone login works.
- **Sharing:** `release/KabadiwalaConnect-most-phones.apk` (34.6 MB) and `...-old-phones.apk` (28.7 MB) can be sent by WhatsApp or Drive and installed like any app.
- **Updates over the internet:** version **1.0.0** is uploaded to **Firebase App Distribution**. Testers added in the Firebase console get an email and can install and update it without cables.
- After installing, the app needs **no cable and no laptop**: it talks to Firebase over the internet, and works offline in between.

## 7. Testing

Every important piece of logic has automatic tests that run with `flutter test`. Current status: **136 tests, all passing**, and `flutter analyze` reports no issues.

| What is tested | Examples |
|---|---|
| Theme | Buttons are at least 56 dp |
| Language files | All 12 languages have exactly the same 123 sentences as Hindi, with the same {slots}; every language name is different |
| Language screen | First page has exactly 4 choices; More shows all 12; Urdu turns the layout right-to-left |
| AI recogniser | labels.txt is read correctly; centre crop is exact; suggestion only at 70%+; OTHER is never suggested; several photos are averaged; the suggested tile comes first and is spoken |
| Real data | Real recyclers carry the CPCB reference and no invented phone or rates; unknown rates are listed after known ones |
| Logout | Asks first; No keeps you logged in; Yes forgets PIN and role but keeps the language; refused while data is unsent |
| Voice | Voice off = nothing automatic but the speaker button still works; switching on says so; Back stops speech |
| Weight | Typing 350 grams and 12 kilo works; grams shown below 1 kg; live price on the Next button |
| Recycler join | Everyone is asked; only linked numbers open recycler mode; others get the apply form; pending vs approved screens |
| Loading screen | Shows logo, name, tagline, ByteStorms; moves on after 2 s; a tap skips it once |
| Safety | Every dangerous material has a card in all 12 languages; first-time card plays before the weight screen; demo reset |
| Big fonts | 7 main screens at 2× text on a 360 dp phone, in Hindi and Tamil, without overflow |
| Sync icon | Shows the number of unsent records |
| Language switching | Tapping a language changes the text instantly and remembers the choice |
| Data models | Every record can be saved and read back unchanged; unknown values fall back safely |
| Seed data | Prices are whole paise and low ≤ high; missing recycler rates are skipped; every row is marked placeholder |
| Offline first run | With no database at all, the app still gets the material list |

| What is tested | Examples |
|---|---|
| NumberSpeaker | 0, 7, 40, 99, 100, 340, 1000, 1340, 99999, 1,50,000 and 1 crore |
| Voice fallback | Recorded clips win; a missing number clip doesn't play half a sentence; Marathi TTS; Hindi fallback |
| Onboarding | Fresh install → language (tap twice) → login; PIN survives a restart and only the right PIN unlocks; recycler mode only for invited numbers |
| Home | 4 tiles, each at least 56 dp, instruction spoken on open |
| Price | 1.7 kg × Rs 180–220 = about Rs 340; rounding to Rs 10; no paisa lost; 500 kg lots stay exact |
| Sell flow | Category → weight (+½) → "about Rs 300" spoken, with no typing at all |
| Trend | Up above +5%, down below −5%, exactly ±5% counts as flat |
| Ranking | Unauthorised and missing rates hidden; formula checked; 25 km cap; ties stable; top 5 only |
| Hash and QR | Same input = same hash; changing **any** field changes it; tampered weight or price is caught; random QR codes rejected |
| Warning flags | Weight off by more than 10%; price outside 0.7×–1.3× of the board |
| Recycler screens | Genuine QR shows green; tampered shows red; another recycler's deal cannot be confirmed |
| Ledger | Paise totals; due shown separately; day starts at local midnight; week starts Monday; no double counting |

### App size

**Honest note:** adding the on-phone AI (about 7 MB of TensorFlow Lite code plus the 2.7 MB model) pushed the main APK over the target. Shrinking it is planned for the polish phase.

The plan's target is a release app under **30 MB**, so it downloads quickly on a cheap data pack. We build one APK per phone processor type ("split per ABI"), so each phone downloads only what it needs:

| APK | For which phones | Size |
|---|---|---|
| `app-armeabi-v7a-release.apk` | Older and very cheap 32-bit phones | **28.7 MB** ✓ |
| `app-arm64-v8a-release.apk` | Most phones from the last ~7 years | **34.6 MB** ! |
| `app-x86_64-release.apk` | Emulators / Chromebooks | 38.9 MB |

**Why the main version is above 30 MB, honestly:** the on-phone AI (4.3 MB engine + 2.7 MB model) and the **offline** QR scanner (5.8 MB) live inside the app. We could save 5.8 MB by letting Google Play services download the scanner later, but then a new recycler phone could not scan QR codes offline until it had been online once, which breaks our "works offline" rule. We chose offline reliability over the last 4.6 MB.

## 8. Data honesty

- **Recyclers** are real, from the official CPCB list (2023); no phone numbers or rates were invented. One clearly named "DEMO Recycler (test account - not real)" exists only for the two-phone demo.

- **Prices** are published web rates with a source on every row, except motors (placeholder). They still need checking with 2–3 local dealers (our field study).
- **Materials** list and **training photos**: team choices and free Wikimedia photos with full credits.
- **Translations** were written by the developers and must be checked by native speakers.
- We never invent real business names or phone numbers.
- The QR hash proves a record **was not changed after it was created**. It is not a legal signature, and we say so in the app.

## 9. Words used in this document

| Word | Meaning |
|---|---|
| APK | The installable file of an Android app |
| Firestore | Google's online database that also works offline |
| Offline-first | The app works fully without internet and syncs later |
| Seed data | A built-in copy of the price/recycler list for the very first run |
| Hash (SHA-256) | A "fingerprint" of data: change one letter and the fingerprint changes completely |
| QR code | A square barcode a phone camera can read |
| TTS | Text-to-speech: the phone reads text aloud |
| Paise | 1/100 of a rupee; we store money in paise to avoid rounding errors |
| UUID | A random unique ID made on the phone |
| Security rules | Server-side rules that decide who may read or change each record |
| Git commit | A saved snapshot of the code we can always go back to |
| Haversine | A formula for the straight-line distance between two GPS points on the Earth |
| Ledger | A record of money earned, paid and still owed |
| Thumbnail | A small, low-resolution copy of a photo |
| AI model | A small program that learned from example photos to recognise objects |
| Training / held-out photos | Photos used to teach the AI / photos kept aside to test it honestly |
| Heat map | A picture showing which part of a photo the AI used to decide |
| CPCB | Central Pollution Control Board, which publishes the list of authorised recyclers |

## 10. What's next (after the hackathon)

| Item | Why |
|---|---|
| Field study in Roorkee and Haridwar | Check prices with 2–3 local dealers; replace the motor placeholder; real collector usability test (can a non-reader finish a sale?) |
| Native-speaker check of all translations | The developers wrote them; Garhwali is still a Hindi copy |

| Item | Why |
|---|---|
| Recorded voice clips for Garhwali and Bhojpuri | Android has no voice for them; the app is ready to play recorded clips |
| More training photos for batteries and motors | The AI's weakest classes (57% and 50%) |
| Contact the 8 authorised recyclers | Get verified phone numbers and rates; invite them to join through the app |
| Admin screen inside the app | So prices and recycler approvals don't need the Mac scripts |
| Google Play Store | Developer account, privacy policy, store listing; the app is already signed for it |